CRYPTOLOGY (CYE 5101)

The Hardest Questions, Cipher by Cipher

Challenge set with full worked solutions at the end | Companion to the Mid-Sem Study Notes

55 questions across 10 topics (at least five per cipher). Every question is a step harder than a typical exam question, and the solutions show the full working. All numbers and ciphertexts in this document were generated and checked by code, so you can trust the answers.

Rating	Meaning
★★★☆☆	Exam level: if you cannot do these, revise the notes first
★★★★☆	Hard: the twist or the arithmetic needs care
★★★★★	Cream-of-the-class: needs insight, proof or a non-obvious idea

How to use this set

1. Cover Part B. Attempt a question on paper with a timer: about **1.2 minutes per mark** is the real exam pace.
2. Write the formula, substitute, reduce mod n, state the answer. Method marks matter.
3. Only then open the solution. If you missed it, redo the question the next day from a blank page.
4. State the attack model (ciphertext-only, known-plaintext, chosen-plaintext) whenever you do cryptanalysis.

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1	Atbash and ROT13	AT1, AT2, AT3, AT4, AT5
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3	Multiplicative Cipher	ML1, ML2, ML3, ML4, ML5
4	Affine Cipher	AF1, AF2, AF3, AF4, AF5, AF6
5	Hill Cipher	HL1, HL2, HL3, HL4, HL5, HL6
6	Vigenère (Polyalphabetic) Cipher	VG1, VG2, VG3, VG4, VG5
7	Columnar Transposition	TR1, TR2, TR3, TR4, TR5
8	Rail Fence Cipher	RF1, RF2, RF3, RF4, RF5
9	One-Time Pad (Vernam)	OT1, OT2, OT3, OT4, OT5
10	DES, 2DES and 3DES	DE1, DE2, DE3, DE4, DE5, DE6, DE7, DE8

Quick Formula Reminder

Cipher	Encrypt	Decrypt	Key space
Shift	$C = P + k$	$P = C - k$	26
Multiplicative	$C = kP$	$P = k^{-1}C$	12 (gcd(k,26)=1)
Affine	$C = aP + b$	$P = a^{-1}(C - b)$	312
Atbash / ROT13	$C = 25 - P$ / $C = P + 13$	same operation	1
Hill	$C = KP$	$P = K^{-1}C$	157,248 (2×2)
Vigenère	$C_i = P_i + K_i$	$P_i = C_i - K_i$	26^m
OTP	$C = P + K$ or $P \oplus K$	$P = C - K$ or $C \oplus K$	26^n
Columnar / rail fence	rows then key-ordered columns / zig-zag	reverse	n! / ~n
DES	$R_i = L_{i-1} \oplus f(R_{i-1}, K_i)$	subkeys reversed	2^{56}

Toolbox

Inverse mod n: extended Euclid. Valid multipliers mod 26: 1, 3, 5, 7, 9, 11, 15, 17, 19, 21, 23, 25 with inverses 1, 9, 21, 15, 3, 19, 7, 23, 11, 5, 17, 25.

Solve $ax \equiv b \pmod n$: $g = \gcd(a, n)$; solvable iff g divides b; then g solutions.

Affine KPA: $a = (c_1 - c_2)(p_1 - p_2)^{-1}$, $b = c_1 - a \cdot p_1$.

Hill KPA: $K = C \cdot P^{-1}$ (columns are digraphs).

S-box: row = outer bits, column = middle four bits.

PART A: Questions

1. Atbash and ROT13

Keyless (or fixed-key) ciphers. They look trivial, but composition and algebra questions are where marks are lost.

AT1 ★★★☆☆ 5 marks

Show that Atbash is a special case of the affine cipher and give its key (a, b). Prove that it is self-inverse. Encrypt **CRYPTOGRAPHY** and verify by decrypting.

AT2 ★★★★☆ 6 marks

A clerk applies **Atbash first and then ROT13** to every message. (a) Write the combined cipher as a single affine key. (b) Encrypt **ZEBRA**. (c) Is the combined cipher self-inverse? (d) Which letters does it leave unchanged?

AT3 ★★★☆☆ 5 marks

A message goes through ROT13 **seven times** and Atbash **four times** (in any order, interleaved). What is the net effect? Justify using group properties. Then say what happens if the number of ROT13 passes is 8.

AT4 ★★★★☆ 6 marks

Prove that, mod 26, the composition of Atbash and any shift by k is again an Atbash-like involution, and find how many such involutions of the form $x \to b - x$ exist. Decrypt **WKLV** if it was produced by $x \to 7 - x$.

AT5 ★★★★★ 10 marks

Kerckhoff and key space. (a) ROT13 and Atbash have a key space of 1. Explain, using Kerckhoff's principle, why neither is a cipher in the real sense. (b) An engineer "improves" ROT13 by chaining ROT13 five times with Atbash in between. Show the resulting scheme still has a key space of 1 and give the affine form. (c) Which of the 312 affine keys include ROT13 and Atbash as special cases?

2. Shift (Caesar) Cipher

$C = (P + k) \bmod 26$, $P = (C - k) \bmod 26$.

SH1 ★★★★☆ 8 marks

Ciphertext-only attack with frequency analysis. The following was produced by a shift cipher: **ESPDP NCPEX PPETY RHTWW MPSPW OYPLC ESPZW OMCTO RPHSP CPESP CTGPC MPYOD**.

The most frequent ciphertext letters are P (15 times), C (6), E (5). (a) Assuming the top letter is E, find k. (b) Decrypt the message. (c) Why would this method fail on a 10-letter ciphertext?

SH2 ★★★☆☆ 6 marks

A careless operator encrypts every message **twice**, first with k_1 and then with k_2. (a) What does the attacker see when $k_1 = 7$ and $k_2 = 19$? (b) The operator then uses **three** passes with keys 11, 15 and 9. Encrypt **CRYPTOLOGY** and state the equivalent single key. (c) Does multiple encryption enlarge the key space? Relate this to why 2DES is not 112-bit secure.

SH3 ★★★★★ 10 marks

A shift cipher is defined on a 27-symbol alphabet: A–Z are 0–25 and the underscore _ (word space) is 26; all arithmetic is mod 27. (a) What is the key space? (b) The ciphertext **FYYMTUMTGHHG** was produced with key k = 20. Decrypt it. (c) Is "ROT13" (k = 13) still self-inverse in this alphabet? Which key is self-inverse? (d) Which keys encrypt the space to itself?

SH4 ★★★★★ 10 marks

Perfect secrecy of the shift cipher (proof question). Messages are single letters, all 26 equally likely or not (any distribution); the key k is chosen uniformly from {0,...,25} and used once. (a) Prove that P(M = m | C = c) = P(M = m) for every m, c. (b) Show by counting that the same property fails for two-letter messages encrypted with one key k.

SH5 ★★★★☆ 8 marks

Pattern-based crib attack. The ciphertext is **LWVWB IBBIK SBPMK IABTM CVBQT BPMAQ OVIT** (a shift cipher). You know the plaintext contains the word **ATTACK**. (a) Which ciphertext substring has the letter pattern of ATTACK (letters 1 and 4 equal, letters 2 and 3 equal)? (b) Find k. (c) Decrypt.

3. Multiplicative Cipher

$C = (k \cdot P) \bmod 26$, $P = (k^{-1} \cdot C) \bmod 26$, valid k: 1, 3, 5, 7, 9, 11, 15, 17, 19, 21, 23, 25.

ML1 ★★★☆☆ 6 marks

A multiplicative cipher encrypts the first plaintext letter **T** as **B**. (a) Find the key. (b) Decrypt the full ciphertext **BSACIYFG**. (c) Write the decryption function.

ML2 ★★★★★ 10 marks

Ambiguous known-plaintext. You learn that plaintext **E** is encrypted to **Y**. (a) List every integer k (0–25) with $4k \equiv 24 \pmod{26}$. (b) Which of them are valid multiplicative keys? (c) State the key and decrypt the ciphertext **YAEYB**. (d) Explain, via CRT, why exactly one of the candidates survives.

ML3 ★★★★☆ 8 marks

Fixed points. (a) Prove that for **every** valid multiplicative key, A and N are encrypted to themselves. (b) Show that, for every valid key other than k = 1, those are the **only** two fixed letters. (c) Explain how this weakness helps an attacker.

ML4 ★★★★☆ 8 marks

Composition. Encrypt with $k_1 = 7$ and then $k_2 = 11$. (a) Find the single equivalent key. (b) Find its inverse. (c) Encrypt **ZEBRA**. (d) Find a pair of valid keys (k_1, k_2) that compose to the identity and explain why the key space is a group.

ML5 ★★★★★ 10 marks

Generalisation to alphabet size 36 (A–Z = 0–25, 0–9 = 26–35). (a) How many valid multiplicative keys exist? (b) Find $5^{-1} \bmod 36$ using the extended Euclidean algorithm. (c) A ciphertext symbol with index 31 was encrypted with k = 5. Decrypt it. (d) Is k = 9 valid? How many keys for alphabet size 91?

4. Affine Cipher

$C = (a \cdot P + b) \bmod 26$, $P = a^{-1}(C - b) \bmod 26$, key space $12 \times 26 = 312$.

AF1 ★★★★★ 10 marks

Known-plaintext attack when $p_1 - p_2$ is **not** invertible. (a) The plaintext **BD** encrypts to **KY**. Find all keys. (b) The plaintext **AN** encrypts to **UH**. Find all keys and explain what is going on.

AF2 ★★★★★ 10 marks

Ciphertext-only affine attack: **DVGDG YDGZB YSVGZ DVGDG MWVGT YDEJY DVGDG YDAZQ ELDVG YDIBG ZDYGN GTCYG UGYDG T**.

Letter counts: G appears 16 times, D appears 13 times. Assume G is the encryption of E and D the encryption of T. (a) Set up and solve for (a, b). (b) Verify gcd(a, 26) = 1. (c) Decrypt the first word. (d) What goes wrong if your assumption is wrong?

AF3 ★★★★☆ 8 marks

Fixed points. (a) For the key (a, b) = (9, 4), find every letter that encrypts to itself. (b) How many of the 312 keys leave **no** letter unchanged? Justify.

AF4 ★★★★☆ 8 marks

Composition. Encrypt first with (5, 8), then with (7, 3). (a) Find the single equivalent key. (b) Find its decryption function. (c) Encrypt **SECURITY** in two ways to confirm. (d) What single key results from encrypting twice with (5, 8)?

AF5 ★★★★★ 10 marks

Affine cipher over a 27-symbol alphabet (A–Z = 0–25, _ = 26, all arithmetic mod 27). (a) Compute the key space. (b) The plaintext **ME** encrypts to **E_**. Find (a, b). (c) Why did a = 3 never appear among your valid multipliers? Give all valid multipliers.

AF6 ★★★★★ 10 marks

Self-inverse keys. (a) Find **all** affine keys (a, b) mod 26 for which encryption and decryption are the same function. (b) How many are there? (c) Of these, how many leave no letter fixed? Name three famous examples.

5. Hill Cipher

$C = K \cdot P \bmod 26$, $P = K^{-1} \cdot C \bmod 26$; K valid iff gcd(det K, 26) = 1.

HL1 ★★★★★ 12 marks

3×3 Hill cipher with K = [[6, 24, 1], [13, 16, 10], [20, 17, 15]]. (a) Encrypt **ACT**. (b) Compute det K mod 26 and decide if K is valid. (c) Compute K^{-1} mod 26. (d) Decrypt your answer to (a) with K^{-1}.

HL2 ★★★★★ 10 marks

Known-plaintext attack on a 2×2 Hill cipher. The plaintext **HELP** encrypts to **HIAT**. (a) Write P and C as matrices whose columns are the plaintext and ciphertext digraphs. (b) Recover $K = C \cdot P^{-1}$. (c) Verify with a third message **GO** → ?.

HL3 ★★★★★ 12 marks

Counting Hill keys. (a) How many invertible 2×2 matrices mod 26 are there? (b) How many have determinant exactly 1 (mod 26)? Use CRT ($Z_{26} \cong Z_2 \times Z_{13}$) and justify each count. (c) Compare with the affine key space and say whether the larger key space makes Hill more secure.

HL4 ★★★★★ 10 marks

Involutory keys. (a) Show that K = [[3, 2], [9, 23]] satisfies $K^2 \equiv I \pmod{26}$ and so equals its own inverse. (b) Encrypt **HI** twice with K and explain. (c) Derive the general condition for a 2×2 key to be involutory and find another one with a = 5.

HL5 ★★★★☆ 8 marks

Chosen-plaintext attack. You may submit any text to the encryption oracle of an unknown 2×2 Hill cipher. You submit **BA** and get **DC**, then **AB** and get **DF**. (a) Recover K. (b) How many letters must you submit for an n×n key? (c) Show that the Hill cipher always maps **AA** to **AA**. (d) Decrypt **HIAT**.

HL6 ★★★★☆ 10 marks

Hill decryption. K = [[5, 8], [17, 3]] and the ciphertext is **OIQVOIWFVDSR**. (a) Show that K is valid and compute K^{-1}. (b) Decrypt. (c) Explain what padding would be needed if the plaintext had 11 letters.

6. Vigenère (Polyalphabetic) Cipher

$C_i = (P_i + K_{i \bmod m}) \bmod 26$, $P_i = (C_i - K_{i \bmod m}) \bmod 26$.

VG1 ★★★★★ 12 marks

Kasiski examination. The ciphertext (spaces removed) is **ELQQEZARZRHSHSEELQFVGIDOAOXTSSTIXROPCABQELQVVWP**.

(a) List the repeated trigrams and the distances between their occurrences. (b) Use the GCD to estimate the key length. (c) The attacker also knows that the plaintext begins with **THE**. Recover the key and decrypt.

VG2 ★★★★☆ 10 marks

Known-plaintext attack. You intercept **VPTZI TTMIJ SUGQH IYIKM SBRUG ZIOIF CSIYI V** (a Vigenère cipher with an unknown key of unknown length) and learn that the message begins with **THESECRET**. (a) Find the key stream for the first nine letters. (b) Recognise the repetition, deduce the key and its length. (c) Decrypt everything. (d) How would a known plaintext shorter than the key length affect the attack?

VG3 ★★★★★ 12 marks

Frequency analysis per column. The ciphertext is **DVOGO JOBOZ OJOBM FOSZS BGOAO FQSNS FSBMO JOBSB QKRSX HRSGS KHRSB UBSGS FSXQY ZNSBH RSXSH HGSOY OBN** and Kasiski suggests a key length of 2. (a) Split into the two columns (odd and even positions). (b) The most frequent letters are column 1: O (13), B (5), R (4); column 2: S (16), B (5), H (4). Assume each column's top letter encrypts E and find the key. (c) Decrypt. (d) Why does this attack need a long ciphertext?

VG4 ★★★★★ 10 marks

Index of coincidence. $IC = \Sigma\, f_i(f_i - 1) / (N(N - 1))$, where f_i is the count of letter i. English text has IC ≈ 0.066; uniformly random letters give 1/26 ≈ 0.038. For the two ciphertexts below compute the IC and decide which is monoalphabetic and which polyalphabetic.

Text 1: NMFAJ FQBFD XYMTZ LMYYM FYYMJ GJXYB FDYTQ JFWSH WDUYT LWFUM DNXYT GWJFP XTRJT KYMJH NUMJW XDTZW XJQKF SIYMJ SFXPB MDYMJ DKFNQ JIYMJ BJFPJ XYXHM JRJXG WJFPV ZNHPQ D

Text 2: TLMJR LPIOL DXTCH RLFHU LXFVR MIEHJ LCFCY PEDBP CCBHB RVMDU JMEHB MVQOX DSYSB QXTSP TTTSE DCAIE DIXTN YHFVR YEEYJ SCFVR JJMWY PHFVR HIMYR DXEQU PQQGO CIMYD FMOYY J

VG5 ★★★★★ 10 marks

Effective key space. A Vigenère key of length m that is a repetition of a shorter key gives the same cipher as the shorter key (e.g. KEYKEY ≡ KEY). (a) How many keys of length 4 are **not** repetitions of a shorter key? (b) The same for length 6. (c) What is the effective key space for all keys of length at most 6?

7. Columnar Transposition

Write by rows, read by columns in key order. Irregular (unpadded) rows make decryption harder.

TR1 ★★★★★ 12 marks

Irregular columnar transposition, no padding. The key is **ZEBRAS** and the ciphertext is **EVLNACDTESEAROFODEECWIREE** (25 letters). (a) Number the key columns. (b) Find how many letters each column holds. (c) Split the ciphertext into columns and decrypt.

TR2 ★★★★☆ 10 marks

Double transposition. Plaintext **MEETATTHEBRIDGEX**, first key **3142**, second key **2413** (digit = read order). (a) Encrypt. (b) Describe how to decrypt. (c) Why is double transposition much harder to break than a single transposition?

TR3 ★★★★☆ 8 marks

Unknown number of columns. The ciphertext **MEERE AOIET LDTTD GMHBE** was produced by a simple columnar transposition (key 12345…, i.e. columns read left to right) with an unknown number of columns n. (a) Which values of n are possible? (b) Try them and recover the plaintext. (c) How would the shape of the ciphertext give you the answer faster?

TR4 ★★★★☆ 8 marks

Key-space arithmetic. (a) A columnar key with 10 columns has how many possible orderings? (b) Double transposition with two independent 10-column keys? (c) An attacker tests 10^6 keys per second: time for each? (d) If the number of columns is unknown but at most 10, how many keys must be considered for a single transposition?

TR5 ★★★★☆ 10 marks

Block permutation cipher. Key π = (4 1 6 2 5 3): ciphertext letter j of each 6-letter block is plaintext letter $\pi(j)$ of that block. (a) Encrypt **ATTACKATDAWN**. (b) Find π^{-1}. (c) Decrypt **AAKTCTAANTWD**.

8. Rail Fence Cipher

Zig-zag over d rails; key = d. Period of the pattern = 2(d − 1).

RF1 ★★★★★ 12 marks

Depth 5 decryption. The ciphertext **DELSEHELOATFTAAFCLEDSWTEENTH** (28 letters) was produced by a rail fence cipher with depth 5. (a) Work out how many letters lie on each rail. (b) Split the ciphertext into rails. (c) Read the zig-zag to recover the plaintext.

RF2 ★★★★☆ 10 marks

Unknown depth. The ciphertext is **MALGOEETODDEONEMTEBIANTHRT**. It is known to be a rail fence cipher with depth between 2 and 6 and to begin (in plaintext) with MEET. (a) Decrypt for every depth and show the first 12 letters. (b) Find the depth and the message. (c) Give a quick test that rules out most depths using the first plaintext letters.

RF3 ★★★★☆ 8 marks

Rail lengths. A 26-letter message is enciphered with depth 5. (a) How many letters fall on each rail? (b) General formula for the number of letters on the first and last rail of an n-letter message with depth d. (c) For n = 26 and d = 5, which rail has the most letters?

RF4 ★★★★★ 10 marks

Product cipher. A message is first Caesar-shifted by +3 and then rail-fenced with depth 3. The ciphertext is **HBDDZQPDWFVWDQHWNG**. (a) Decrypt. (b) Does the order of the two operations matter? (c) Which of Shannon's properties does each stage contribute?

RF5 ★★★★★ 10 marks

Distinct keys. For the 12-letter message **ABCDEFGHIJKL** (all letters different), (a) for which depths d does the rail fence leave the text unchanged? (b) How many distinct ciphertexts are possible for d = 2, 3, ..., 12? (c) Is the rail fence key space ever more than about n? Give the exact count for n = 12.

9. One-Time Pad (Vernam)

$C_i = (P_i + K_i) \bmod 26$ or $C = P \oplus K$. Perfect secrecy needs a random, equal-length, single-use key.

OT1 ★★★★★ 12 marks

Two-time pad. Two messages of equal length were encrypted with the **same** letter-OTP key: C_1 = **PUMMYPEORJDZ**, C_2 = **EEKMBPJSMEDQ**. You know P_1 begins with **SEND** and ends with **NOW**. (a) Compute $C_1 - C_2$. (b) Recover the key letters you can. (c) Recover the matching letters of P_2 and guess the rest. (d) State the lesson.

OT2 ★★★★★ 12 marks

Malleability. A bank sends "PAY 100" encrypted with a byte-wise XOR one-time pad. The ciphertext in hex is **6A8427B139E452**. You do not know the key but know the plaintext format. (a) Which byte do you change and by what value to make it read "PAY 900"? (b) Give the new ciphertext. (c) Show that decryption gives PAY 900. (d) What does this show about confidentiality vs integrity?

OT3 ★★★★★ 12 marks

Why the key must be uniform. Use a 3-letter alphabet {A, B, C} = {0, 1, 2} with shift encryption C = M + K mod 3. Message priors: P(A) = 0.5, P(B) = 0.3, P(C) = 0.2. (a) If K is uniform on {0, 1, 2}, find P(M = A | C = A). (b) If P(K = 0) = 0.6, P(K = 1) = P(K = 2) = 0.2, find P(M | C = A) for all three messages. (c) Conclude.

OT4 ★★★☆☆ 6 marks

Practicality. A secure video call uses 2 Mbit/s for 90 minutes. (a) How many bytes of one-time-pad key are consumed? (b) If the key is stored on a 128 GB drive, how many such calls can be protected (decimal GB)? (c) Why does this make OTP unpopular?

OT5 ★★★★☆ 8 marks

Deniability. The letter-OTP ciphertext **EYSXG** decrypts to **RIVER** with key **NQXTP**. (a) What key makes the same ciphertext decrypt to **ARENA**? (b) Why can no one tell which plaintext is real? (c) Link this to the Caesar EVIRE question.

10. DES, 2DES and 3DES

All computations below use the standard DES tables given in the notes (Chapter 9). The sample key is 133457799BBCDFF1 and the plaintext is 0123456789ABCDEF.

DE1 ★★★★★ 15 marks

One full DES round (round 2). After round 1, L_1 = **1111 0000 1010 1010 1111 0000 1010 1010** and R_1 = **1110 1111 0100 1010 0110 0101 0100 0100** (32 bits each). The round-2 subkey is K_2 = **011110 011010 111011 011001 110110 111100 100111 100101**. (a) Expand R_1 with the E-box to 48 bits. (b) XOR with K_2. (c) Pass each 6-bit group through its S-box. (d) Apply the P-box and compute $R_2 = L_1 \oplus f(R_1, K_2)$. Give L_2.

DE2 ★★★★★ 12 marks

Key schedule. After PC-1 the halves are C_0 = **1111000011001100101010101111** and D_0 = **0101010101100110011110001111** (28 bits each). (a) State the number of left shifts in rounds 1 and 2. (b) Compute C_2 and D_2. (c) Apply PC-2 to find K_2 (use the PC-2 table). (d) PC-2 uses only 48 of the 56 bits. Which 8 positions are dropped in every round, and is any key bit therefore unused overall?

DE3 ★★★★☆ 10 marks

S-boxes and P-box. The 48-bit input to the S-boxes is **101011 000111 010010 111100 001001 110110 100101 011010**. (a) Compute the eight 4-bit outputs S1…S8. (b) Concatenate to 32 bits. (c) Apply the P-box to get the round function output. (d) State the number of rows, columns and bits per S-box entry.

DE4 ★★★★★ 12 marks

Feistel network with a non-invertible function. Block = 8 bits (L_0 = 1011, R_0 = 0110), two rounds, subkeys K_1 = 1100 and K_2 = 0101, round function f(R, K) = (R AND K) ⊕ (R rotated left by 1). Rounds: $L_i = R_{i-1}$, $R_i = L_{i-1} \oplus f(R_{i-1}, K_i)$; after round 2 swap the halves. (a) Encrypt. (b) Decrypt using the same structure with the subkeys reversed. (c) Show that f is not invertible, yet decryption works.

DE5 ★★★★★ 12 marks

Complementation property. (a) State and **prove** that DES_{~K}(~P) = ~DES_K(P), where ~ is the bitwise complement. (b) The standard vector gives DES_K(0123456789ABCDEF) = 85E813540F0AB405 for K = 133457799BBCDFF1. Without running DES, what is the encryption of FEDCBA9876543210 under the key ECCBA8866443200E? (c) How does the property help a brute-force attacker?

DE6 ★★★★★ 12 marks

Meet-in-the-middle on a toy 2E. A toy block cipher on 4-bit blocks is E_k(x) = (3x + k) mod 16 with a 3-bit key k ∈ {0, …,7}. Double encryption is C = E_{k2}(E_{k1}(P)). You know two plaintext/ciphertext pairs: (7 → 0) and (2 → 3). (a) How many key pairs would brute force test? (b) Build the forward table E_k(P) for k = 0…7 and the backward table

D_k(C). (c) List the matches. (d) Filter the candidates with the second pair. Is the key pair unique? Explain. (e) Count the work compared with brute force.

DE7 ★★★★☆ 10 marks

Complexity of attacks. Assume 10^9 DES operations per second. (a) Time for a brute-force search of DES (average case: half the key space). (b) Operations, time and memory (8 bytes per table entry) for MITM on 2DES. (c) Operations for MITM on three-key 3DES (2^{112}) and the time in years. (d) Why is two-key 3DES (112-bit key) still considered better than 2DES?

DE8 ★★★★☆ 10 marks

Initial and final permutation. (a) Apply the initial permutation IP to M = 0123456789ABCDEF (write M in binary first) and give L_0 and R_0 in hex. (b) Which input bit goes to output bit 1 and which to output bit 64? (c) Apply the final permutation FP to the 64-bit IP output and show you recover M. (d) Explain why IP and FP have no cryptographic strength.

PART B: Worked Solutions

Each solution gives the method first, then the arithmetic, then the final answer in bold.

1. Atbash and ROT13

Solution AT1

Atbash maps $x \to 25 - x = 25x + 25 \pmod{26}$ because $-1 \equiv 25$. So it is the affine cipher with **a = 25, b = 25**.

Self-inverse: applying twice gives $25 - (25 - x) = x$. In affine terms $a^2 = 625 = 24 \cdot 26 + 1 \equiv 1$ and $(a + 1)b = 26 \cdot 25 \equiv 0$.

Encrypt CRYPTOGRAPHY: **XIBKGLTIZKSB**. Applying Atbash again returns CRYPTOGRAPHY. ✓

Solution AT2

(a) $x \to 25 - x \to (25 - x) + 13 = 38 - x \equiv 12 - x \equiv 25x + 12$. **Key (a, b) = (25, 12).**

(b) ZEBRA → Atbash AVYIZ → ROT13 NILVM. Direct formula gives NILVM. **NILVM**

(c) Yes. For a = 25 we have $a^2 \equiv 1$ and $(a + 1)b = 26b \equiv 0$ for every b, so every key with a = 25 is an involution.

(d) Fixed points: $12 - x \equiv x \Rightarrow 2x \equiv 12 \pmod{26} \Rightarrow x \equiv 6 \pmod{13} \Rightarrow x = 6$ or 19. **G and T are unchanged.** (Check: 12 − 6 = 6, 12 − 19 = −7 ≡ 19.)

Solution AT3

ROT13 and Atbash are both involutions (self-inverse), so applying either an even number of times gives the identity.

Do they commute? ROT13∘Atbash: $25 - x + 13 = 38 - x$; Atbash∘ROT13: $25 - (x + 13) = 12 - x \equiv 38 - x \pmod{26}$ because 38 − 12 = 26. Yes, they commute, so the order of interleaving does not matter.

Seven ROT13 passes = one ROT13 (six cancel in pairs). Four Atbash passes = identity. **Net effect = ROT13.** With eight ROT13 passes the net effect would be the identity (no encryption at all).

Solution AT4

Shift then Atbash: $25 - (x + k) = (25 - k) - x$. Atbash then shift: $(25 - x) + k = (25 + k) - x$. Both have the form $x \to b - x$.

Any map $x \to b - x$ is an involution because $b - (b - x) = x$. There is one for each $b \in \{0, \ldots, 25\}$, so **26 involutions** of this form (they are the 26 affine keys with a = 25).

Decrypt WKLV with $x \to 7 - x$ (self-inverse, so apply the same map): W=22→11=L, K=10→23=X, L=11→22=W, V=21→12=M. **LXWM**

Solution AT5

(a) Kerckhoff: security must depend only on the key, assuming the attacker knows the algorithm. For ROT13/Atbash there is no key at all, so once the algorithm is known (it is) everything is public. They are encodings that hide text from casual readers only.

(b) The scheme is ROT13, Atbash, ROT13, Atbash, ROT13, Atbash, ROT13, Atbash, ROT13: five ROT13 passes and four Atbash passes. ROT13 and Atbash commute and are both involutions, so five ROT13 passes collapse to one ROT13 and four Atbash passes to the identity. The net is plain ROT13: $x \to x + 13$, affine (a, b) = (1, 13). It is still **one fixed transformation with key space 1**; the extra passes add work, not security.

(c) ROT13 = affine (1, 13); Atbash = affine (25, 25). Both are among the 312 keys, and both are involutions. Neither adds any secrecy because the key is fixed and public.

2. Shift (Caesar) Cipher

Solution SH1

(a) Most frequent ciphertext letter P = 15; assumed plaintext E = 4. k = 15 − 4 = **11**.

(b) Subtract 11 from every letter. Plaintext: **THE SECRET MEETING WILL BE HELD NEAR THE OLD BRIDGE WHERE THE RIVER BENDS**.

(c) Frequency analysis relies on the law of large numbers. In a short ciphertext the letter counts are too random: E may not even be the most common letter. For short messages brute force over the 26 keys (and recognising the correct plaintext) is the reliable method.

Solution SH2

(a) $7 + 19 = 26 \equiv 0 \pmod{26}$. The two passes cancel and **the ciphertext equals the plaintext** (no encryption).

(b) $11 + 15 + 9 = 35 \equiv 9$. Equivalent single key **k = 9**. CRYPTOLOGY → **LAHYCXUXPH**.

(c) No. Shift ciphers are closed under composition (they form a group): E_{k2}(E_{k1}(x)) = E_{k1+k2}(x). The key space stays at 26. DES is believed not to be a group, so 2DES does add something, but the meet-in-the-middle attack still reduces 2DES to about 2^{57} operations instead of 2^{112}.

Solution SH3

(a) Key space = **27** (k = 0 … 26).

(b) Subtract 20 (equivalently add 7) mod 27: **MEET_AT_NOON**.

(c) A shift is self-inverse when $2k \equiv 0 \pmod{27}$. 27 is odd, so 2 is invertible, which forces $k \equiv 0$. Therefore **no non-trivial shift is self-inverse**; $13 + 13 = 26 \not\equiv 0 \pmod{27}$. With k = 13 you would need another pass of k = 14 to undo it.

(d) Space (26) maps to itself when $26 + k \equiv 26$, i.e. **only k = 0**.

Solution SH4

(a) For any m and c there is exactly one key $k = c - m$, so $P(C = c \mid M = m) = P(K = c - m) = 1/26$, independent of m. Then $P(C = c) = \Sigma_m P(m)\cdot(1/26) = 1/26$. Bayes: $P(m \mid c) = P(c \mid m)P(m)/P(c) = (1/26)P(m)/(1/26) =$ **P(m)**. So a single-letter shift cipher with a fresh uniform key has perfect secrecy.

(b) For two-letter messages there are $26^2 = 676$ possible plaintexts but only 26 keys, so for a given ciphertext only 26 of the 676 plaintexts are possible. Any plaintext whose letters do not differ by the same shift as the ciphertext letters has posterior probability 0, while the right family gets 1/26 each. E.g. ciphertext "AB": a plaintext such as "AA" is impossible (it would need k = 0 for the first letter and k = 25 for the second), so the posterior differs from the prior. **Perfect secrecy needs $|K| \geq |M|$** (Shannon).

Solution SH5

(a) ATTACK has the letter pattern A T T A C K: letter 1 equals letter 4 and letter 2 equals letter 3 (pattern xyyxzw). Scan the spaceless ciphertext for six letters with this pattern. The substring **IBBIKS** matches this pattern.

(b) A → I: $k = 8 - 0 =$ **8**. Check $T(19) + 8 = 27 \equiv 1 = B$, $C(2) + 8 = 10 = K$. ✓

(c) Subtract 8 from every letter: **DO NOT ATTACK THE CASTLE UNTIL THE SIGNAL**.

3. Multiplicative Cipher

Solution ML1

(a) T = 19, B = 1. Solve $19k \equiv 1 \pmod{26}$. $19^{-1} = 11$ ($19\cdot11 = 209 = 8\cdot26 + 1$). $k = 11\cdot1 \bmod 26 =$ **11**.

(b) $11^{-1} = 19$ ($11\cdot19 = 209 \equiv 1$). Multiply each ciphertext number by 19: **TEAMWORK**.

(c) $D(y) = 19y \bmod 26$.

Solution ML2

(a) $4k \equiv 24 \pmod{26}$. gcd(4, 26) = 2 divides 24, so there are 2 solutions modulo 26. Divide by 2: $2k \equiv 12 \pmod{13} \Rightarrow k \equiv 6 \pmod{13}$. So **k = 6 or k = 19**.

(b) gcd(6, 26) = 2 → invalid. gcd(19, 26) = 1 → valid. Only **k = 19** is a legal key.

(c) $19^{-1} = 11$. Decrypt YAEYB: multiply each letter number by 11: **EASEL**.

(d) The two solutions differ by 13: $6 \equiv 19 \pmod{13}$ but 6 is even and 19 is odd (they differ mod 2). By CRT, $Z_{26} \cong Z_2 \times Z_{13}$, and a valid key must be odd (invertible mod 2). Exactly one of two solutions that differ by 13 is odd, so exactly one survives.

Solution ML3

(a) A = 0: k·0 = 0. N = 13: k·13 ≡ 13 (mod 26) for odd k because 13(k − 1) = 13·(even) = 26·m ≡ 0. All valid k are odd. So **A and N always map to themselves**.

(b) $kx \equiv x \Leftrightarrow (k-1)x \equiv 0 \pmod{26}$. For odd $k \neq 1$, $k-1$ is even and not a multiple of 13 ($k = 14$ is even, so excluded), hence $\gcd(k-1, 26) = 2$. Number of solutions x in Z_{26} is $\gcd(k-1, 26) = 2$, namely $x = 0$ and $x = 13$.

(c) The attacker knows two letters of the mapping for free, and A and N are never encrypted to anything else; frequency counts of the letters A and N in the ciphertext equal those in the plaintext. The key can be found with just one other known pair.

Solution ML4

(a) k = 7·11 = 77 ≡ 77 − 52 = **25**.

(b) 25·25 = 625 = 24·26 + 1 ≡ 1, so the inverse is 25 (the key 25 is $x \to -x$).

(c) ZEBRA: Z=25→25·25 = 625 ≡ 1 (B), E=4→100 ≡ 22 (W), B=1→25 (Z), R=17→425 ≡ 9 (J), A→A. **BWZJA**

(d) e.g. (3, 9): 27 ≡ 1. Valid keys are the units of Z_{26}; the product of two units is a unit, 1 is the identity, and every unit has an inverse: a group of order 12. Hence double encryption adds no strength.

Solution ML5

(a) $\varphi(36) = 36 \cdot (1-1/2)(1-1/3) =$ **12**.

(b) 36 = 7·5 + 1 ⇒ 1 = 36 − 7·5, so $5^{-1} \equiv -7 \equiv$ **29** (mod 36). Check 5·29 = 145 = 4·36 + 1. ✓

(c) 29·31 = 899 = 24·36 + 35, so plaintext index 35, which is the digit **9**.

(d) gcd(9, 36) = 9 ≠ 1, so k = 9 is invalid. For 91 = 7×13, φ(91) = 6·12 = **72** keys.

4. Affine Cipher

Solution AF1

(a) B=1→K=10, D=3→Y=24. Subtract: 10 − 24 = a(1 − 3) ⇒ −14 ≡ 12 = −2a ⇒ 2a ≡ 14 (mod 26) ⇒ a ≡ 7 (mod 13) ⇒ a = 7 or 20. a = 20 is invalid (even). **a = 7**, b = 10 − 7 = **3**. Check: 7·3 + 3 = 24 = Y. ✓ Key (7, 3), unique.

(b) A = 0 → U = 20, so **b = 20**. N = 13 → H = 7: 13a + 20 ≡ 7 ⇒ 13a ≡ −13 ≡ 13 (mod 26). For odd a, 13a ≡ 13, so **every valid a satisfies it**. Thus 12 keys (a, 20) with $a \in \{1, 3, 5, 7, 9, 11, 15, 17, 19, 21, 23, 25\}$ all fit.

Lesson: $p_1 - p_2 = 13$ is a zero divisor, giving no information about a. Always choose plaintext letters whose difference is coprime to 26 (e.g. an odd difference other than 13), or use a third pair.

Solution AF2

(a) E=4→G=6, T=19→D=3. Subtract: 6 − 3 = a(4 − 19) ⇒ 3 ≡ −15a ≡ 11a (mod 26). $11^{-1} = 19$, so a = 19·3 mod 26 = **5**. b = 6 − 5·4 = **12**.

(b) gcd(5, 26) = 1. ✓

(c) $5^{-1} = 21$; P = 21(C − 12). First word: **THE**; the full plaintext is THE TEST ENDS WHEN THE TEACHER STOPS THE TESTING OF THE STUDENTS EVERY SEMESTER.

(d) If the assumption is wrong the resulting key either has gcd(a, 26) ≠ 1 (reject it) or decrypts the text into gibberish. Then try the next pairing (second-most-frequent letter as E, and so on), up to a handful of attempts. Always verify by decrypting a sentence.

Solution AF3

(a) Solve 9x + 4 ≡ x ⇒ 8x ≡ −4 ≡ 22 (mod 26). gcd(8, 26) = 2 divides 22, so two solutions: 4x ≡ 11 (mod 13); 4^{-1} mod 13 = 10, so x ≡ 110 ≡ 6 (mod 13) ⇒ x = 6 or 19. **G and T are unchanged.** Check: 9·6 + 4 = 58 ≡ 6; 9·19 + 4 = 175 ≡ 19. ✓

(b) (a − 1)x ≡ −b (mod 26). Case a = 1: no solution iff b ≠ 0 → 25 keys. Case a ≠ 1 (11 values, all odd): a − 1 is even and gcd(a − 1, 26) = 2, so a solution exists iff 2 | b. So no fixed point iff b is odd: 13 values → 11·13 = 143 keys. Total = 25 + 143 = **168**.

Solution AF4

(a) $7(5x + 8) + 3 = 35x + 59 \equiv 9x + 7$. **Key (9, 7).**

(b) $9^{-1} = 3$. $D(y) = 3(y - 7) = 3y - 21 \equiv$ **3y + 5**.

(c) Step by step: UCSEPWZY → NRZFEBWP. Single key (9, 7): NRZFEBWP. Equal. ✓

(d) $5(5x + 8) + 8 = 25x + 48 \equiv$ **25x + 22**, key (25, 22), still a single affine key (and an involution since a = 25).

Solution AF5

(a) $\varphi(27) = 27 \cdot (1 - 1/3) = 18$ multipliers (numbers coprime to 3). Key space = 18 × 27 = **486**.

(b) M=12→E=4 and E=4→_=26. Subtract: $12a + b \equiv 4$ and $4a + b \equiv 26 \Rightarrow 8a \equiv -22 \equiv 5 \pmod{27}$. $8^{-1} \bmod 27 = 17$ ($8 \cdot 17 = 136 = 5 \cdot 27 + 1$). $a = 5 \cdot 17 = 85 \equiv$ **4**. $b = 26 - 4 \cdot 4 =$ **10**. Check: $4 \cdot 12 + 10 = 58 \equiv 4$ (E). ✓

(c) Valid multipliers must be coprime to $27 = 3^3$, i.e. not a multiple of 3: {1,2,4,5,7,8,10,11,13,14,16,17,19,20,22,23,25,26}. a = 3 is a multiple of 3 and so not invertible. Note that even numbers ARE allowed here, unlike mod 26.

Solution AF6

(a) Need $E(E(x)) = x$: $a^2x + ab + b \equiv x$ for all $x \Rightarrow a^2 \equiv 1$ and $(a + 1)b \equiv 0 \pmod{26}$. $a^2 \equiv 1 \pmod{26}$ has solutions a = 1, 25 (units mod 2 and mod 13: $a \equiv \pm 1 \bmod 13$ and odd).

a = 1: $2b \equiv 0 \Rightarrow b = 0$ or 13. a = 25: $26b \equiv 0$ holds for every $b \Rightarrow b = 0 \ldots 25$.

(b) $2 + 26 =$ **28 keys**.

(c) a = 1: b = 13 has no fixed point (b = 0 is the identity). a = 25: fixed points solve $24x \equiv -b$, i.e. need $\gcd(24, 26) = 2 \mid b$; no fixed point iff b odd (13 keys). So **14** keys are fixed-point free.

Examples: identity (1, 0), ROT13 (1, 13), Atbash (25, 25).

5. Hill Cipher

Solution HL1

(a) ACT = (0, 2, 19). Row 1: $6 \cdot 0 + 24 \cdot 2 + 1 \cdot 19 = 67 \equiv 15$ (P). Row 2: $13 \cdot 0 + 16 \cdot 2 + 10 \cdot 19 = 222 \equiv 14$ (O). Row 3: $20 \cdot 0 + 17 \cdot 2 + 15 \cdot 19 = 319 \equiv 7$ (H). **POH**

(b) $\det = 6(16 \cdot 15 - 10 \cdot 17) - 24(13 \cdot 15 - 10 \cdot 20) + 1(13 \cdot 17 - 16 \cdot 20) = 6(70) - 24(-5) + (-99) = 420 + 120 - 99 = 441 \equiv 25 \pmod{26}$. $\gcd(25, 26) = 1 \rightarrow$ **valid**. $25^{-1} = 25$.

(c) $K^{-1} = 25 \cdot \text{adj}(K) \bmod 26 =$ **[[8, 5, 10], [21, 8, 21], [21, 12, 8]]**.

(d) Applying K^{-1} to (15, 14, 7) returns (0, 2, 19) = **ACT**. ✓

Solution HL2

(a) HE = (7, 4), LP = (11, 15). P = [[7, 11], [4, 15]]. HI = (7, 8), AT = (0, 19). C = [[7, 0], [8, 19]].

(b) $\det P = 7 \cdot 15 - 11 \cdot 4 = 61 \equiv 9$; $9^{-1} = 3$. $P^{-1} = 3 \cdot [[15, -11], [-4, 7]] = [[45, -33], [-12, 21]] \equiv [[19, 19], [14, 21]]$. $K = C \cdot P^{-1} \bmod 26 =$ **[[3, 3], [2, 5]]**

(c) GO = (6, 14). $K \cdot (6, 14) = (3 \cdot 6 + 3 \cdot 14, 2 \cdot 6 + 5 \cdot 14) = (60, 82) \equiv (8, 4) =$ **IE**.

Solution HL3

(a) $|GL_2(Z_p)| = (p^2 - 1)(p^2 - p)$. For p = 2: $3 \cdot 2 = 6$. For p = 13: $168 \cdot 156 = 26{,}208$. By CRT: $6 \cdot 26{,}208 =$ **157,248**.

(b) $|SL_2(Z_p)| = p(p^2 - 1)$. p = 2: $2 \cdot 3 = 6$. p = 13: $13 \cdot 168 = 2{,}184$. Product = $6 \cdot 2{,}184 =$ **13,104** (brute-force check: 13104).

(c) Affine has 312 keys; Hill 2×2 has 157,248. But Hill is linear and falls to a known-plaintext attack with two digraphs: $K = C \cdot P^{-1}$. A large key space is necessary, not sufficient, for security.

Solution HL4

(a) $K^2 = [[3 \cdot 3 + 2 \cdot 9, 3 \cdot 2 + 2 \cdot 23], [9 \cdot 3 + 23 \cdot 9, 9 \cdot 2 + 23 \cdot 23]] = [[27, 52], [234, 547]] \equiv [[1, 0], [0, 1]] = I$. ✓ $\det K = 69 - 18 = 51 \equiv 25 = -1$.

(b) HI = (7, 8): K·(7, 8) = (21 + 16, 63 + 184) = (37, 247) ≡ (11, 13) = LN. Second pass returns HI = HI. Encrypting twice gives the plaintext back.

(c) K = [[a, b], [c, d]] is involutory iff $K^2 = I \Leftrightarrow d = -a$ (when $K \neq \pm I$) and $a^2 + bc \equiv 1 \pmod{26}$. With a = 5: d = −5 ≡ 21 and bc ≡ 1 − 25 = −24 ≡ 2. Choose b = 1, c = 2: **K = [[5, 1], [2, 21]]** (det = 105 − 2 = 103 ≡ 25 = −1, valid). Check K^2: [[25 + 2, 5 + 21], [10 + 42, 2 + 441]] = [[27, 26], [52, 443]] ≡ [[1, 0], [0, 1]]. ✓

Solution HL5

(a) K·(1, 0) = first column of K = (D, C) = (3, 2). K·(0, 1) = second column = (D, F) = (3, 5). **K = [[3, 3], [2, 5]]**.

(b) One block per column: n blocks of n letters = **n^2 letters** (e.g. BAA, ABA, AAB for 3×3: 9 letters).

(c) Linearity: K·0 = 0, so AA → AA (and AAA → AAA for 3×3). This is a visible weakness: padding with A is detectable.

(d) det K = 15 − 6 = 9, $9^{-1} = 3$. $K^{-1} = 3 \cdot [[5, -3], [-2, 3]] = [[15, -9], [-6, 9]] \equiv [[15, 17], [20, 9]]$. Decrypt HIAT: **HELP**.

Solution HL6

(a) det = 15 − 136 = −121 ≡ 9 (mod 26); gcd(9, 26) = 1 → valid. $9^{-1} = 3$. $K^{-1} = 3 \cdot [[3, -8], [-17, 5]] = [[9, -24], [-51, 15]] \equiv$ **[[9, 2], [1, 15]]**.

(b) Multiply each digraph by K^{-1}: **MEETMEATNOON**.

(c) The plaintext length must be a multiple of 2, so add one filler letter (conventionally X) at the end: 11 → 12 letters. The receiver removes it.

6. Vigenère (Polyalphabetic) Cipher

Solution VG1

(a) Repeats: ELQ at positions [0, 15, 40] (distances [15, 40, 25]).

(b) GCD of all distances = **5**, so the key length is 5 (or a factor of the distances; here 5 is the largest consistent candidate).

(c) The first three ciphertext letters ELQ minus T, H, E give the key: E−T = 11 (L), L−H = 4 (E), Q−E = 12 (M). **Key = LEMON**. Plaintext: THE CROW FLEW OVER THE RIVER AND THE FIELD BEYOND THE HILL.

Solution VG2

(a) Subtract plaintext from ciphertext: V−T = C, P−H = I, T−E = P, Z−S = H, I−E = E, T−C = R, T−R = C, M−E = I, I−T = P → key stream **CIPHERCIP**.

(b) The key stream CIPHERCIP repeats after 6 letters, so the key is **CIPHER**, length 6.

(c) Apply the key to the whole ciphertext: **THESECRETCODEISBURIEDUNDERTHEOAKTREE**.

(d) If the known plaintext is shorter than the key you only recover part of the key and must guess or crib the remainder; if it is at least the key length, one full period reveals everything.

Solution VG3

(a) Column 1 = letters 1, 3, 5, …; column 2 = letters 2, 4, 6, …. Each column is a Caesar cipher.

(b) Column 1: O = 14, E = 4 ⇒ $k_1 = 10$ (K). Column 2: S = 18, $k_2 = 14$ (O). **Key = KO**.

(c) Decrypt with the key KO:
THESEVENELEVENCREEPERSEMERGEDEVERYEVENINGWHENTHEWEATHERGREWEVENCOLDERTHENEXTWEEKEND.

(d) Each column has only about N/m letters; for frequency counts to approximate English you need dozens of letters per column. With a longer key and short text, the columns are too short for reliable peaks.

Solution VG4

Counting letters (N = 136 each): IC(Text 1) = **0.0626**, IC(Text 2) = **0.0434**.

Text 1 is close to the English value 0.066, so letter frequencies survive: **monoalphabetic** (it is a Caesar cipher with k = 5). Text 2 is lower, closer to the random value 0.038, so the frequencies have been flattened: **polyalphabetic** (Vigenère, key LEMON).

Use: IC identifies the cipher class, and computing the IC of every m-th column and looking for values near 0.066 gives the key length m.

Solution VG5

(a) Keys of length 4 that repeat a shorter key: periods 1 and 2: $26^2 = 676$ (period 1 is included in period 2). So primitive keys = $26^4 - 26^2 = 456{,}976 - 676 =$ **456,300**.

(b) Length 6: period 2 gives $26^2 = 676$, period 3 gives $26^3 = 17{,}576$, their overlap (period 1) is 26. Union = 676 + 17,576 − 26 = 18,226. Primitive keys = 308,915,776 − 18,226 = **308,897,550** (Mobius formula: $26^6 - 26^3 - 26^2 + 26$).

(c) Primitive counts: m = 1: 26; m = 2: 676 − 26 = 650; m = 3: 17,576 − 26 = 17,550; m = 4: 456,300; m = 5: 11,881,376 − 26 = 11,881,350; m = 6: 308,897,550. Total = **321,253,426**.

7. Columnar Transposition

Solution TR1

(a) Alphabetical order of ZEBRAS: A(1) at position 5, B(2) at 3, E(3) at 2, R(4) at 4, S(5) at 6, Z(6) at 1. Column numbers by position: **6 3 2 4 1 5**.

(b) 25 letters in 6 columns = 4 full rows + 1 extra letter. The extra letter falls in column 1 (position 1, key Z). So column lengths by position: [5, 4, 4, 4, 4, 4].

(c) Reading order is number 1,2,3,4,5,6, i.e. positions 5, 3, 2, 4, 6, 1 with lengths [4, 4, 4, 4, 4, 5]. Cut the ciphertext accordingly: EVLN | ACDT | ESEA | ROFO | DEEC | WIREE. Rebuild row by row: **WEAREDISCOVEREDFLEEATONCE**.

Solution TR2

(a) First pass (4 columns, 4 rows, key 3142): ETBGTHIXMAEDETRE. Second pass with key 2413: **BIERETMEGXDETHAT**.

(b) Undo the second transposition first (with key 2413), then the first (key 3142): decryption reverses the order of operations.

(c) The first pass leaves regular column structure that a single-pass attacker (anagramming) can exploit; the second pass scrambles that structure, so the combined permutation is not a simple columnar one, the number of keys is multiplied, and standard multiple-anagramming attacks do not apply directly.

Solution TR3

(a) The length is 20, so n must divide 20 (for an unpadded complete rectangle): $n \in \{2, 4, 5, 10\}$.

(b) Try n = 4 (columns of 5): reading rows gives gibberish (MALGEODMEITHRETBETDE). Try n = 5 (columns of 4): columns MEER | EAOI | ETLD | TTDG | MHBE. Writing them side by side: row 1 MEETM, row 2 EATTH, row 3 EOLDB, row 4 RIDGE. Read rows: **MEETMEATTHEOLDBRIDGE** = “Meet me at the old bridge”. n = 5.

(c) With a known crib such as MEET at the start, the first letters of the first columns must spell it. For n = 5 the first letters of the columns are M, E, E, T, M, so the first row reads MEETM. Checking the first row of each candidate rectangle is much faster than decrypting all of them.

Solution TR4

(a) 10! = **3,628,800**.

(b) $(10!)^2$ = **13,168,189,440,000 ≈ 1.3×10^{13}**.

(c) Single: $3.63\times10^6 / 10^6 \approx$ **3.6 s**. Double: $1.3168\times10^{13} / 10^6 = 1.3168\times10^7$ s ≈ **152 days**.

(d) Sum of n! for n = 2 … 10 = 2 + 6 + 24 + 120 + 720 + 5,040 + 40,320 + 362,880 + 3,628,800 = **4,037,912**.

Solution TR5

(a) Block 1 ATTACK: letters at positions 4,1,6,2,5,3 = A,A,K,T,C,T → AAKTCT. Block 2 ATDAWN → A,A,N,T,W,D → AANTWD. **AAKTCTAANTWD**.

(b) $\pi(1)=4 \Rightarrow \pi^{-1}(4)=1$; $\pi(2)=1 \Rightarrow \pi^{-1}(1)=2$; $\pi(3)=6 \Rightarrow \pi^{-1}(6)=3$; $\pi(4)=2 \Rightarrow \pi^{-1}(2)=4$; $\pi(5)=5 \Rightarrow \pi^{-1}(5)=5$; $\pi(6)=3 \Rightarrow \pi^{-1}(3)=6$. **$\pi^{-1}$ = (2 4 6 1 5 3)**.

(c) Apply π^{-1} to each block: **ATTACKATDAWN**.

8. Rail Fence Cipher

Solution RF1

(a) Pattern of rails for positions 0, 1, 2, …: 1 2 3 4 5 4 3 2 | 1 2 3 4 5 4 3 2 | … (period 8). For 28 letters (3 full periods + 4): rail 1: **4**, rail 2: **7**, rail 3: **7**, rail 4: **7**, rail 5: **3** (total 28).

(b) Rails: 1: DELS | 2: EHELOAT | 3: FTAAFCL | 4: EDSWTEE | 5: NTH.

(c) Walk down rails 1→5 then up 4→2 repeatedly, taking the next unused letter from each rail: **DEFENDTHEEASTWALLOFTHECASTLE**.

Solution RF2

(a) depth 2: MNAELMGTOEEB; depth 3: MTAOADNDLETO; depth 4: MEETMEATTHEO; depth 5: MOEEHBOEAENI; depth 6: MGONBHIEDOAE.

(b) Only depth 4 gives text beginning MEET. Message: **MEETMEATTHEOLDBRIDGEATNOON**.

(c) Decrypt only the first four letters for each depth: a wrong depth fails to produce the crib within four letters. Structural shortcut: the first rail holds plaintext letters 1, 2d − 1, 4d − 3, …, so the second ciphertext letter must be plaintext letter 2d − 1 (for d = 4 that is letter 7, an A, and the ciphertext does start MA).

Solution RF3

(a) Period 2(d − 1) = 8. 26 = 3·8 + 2. Each full period gives rail 1: 1, rail 2: 2, rail 3: 2, rail 4: 2, rail 5: 1. The leftover 2 letters fall on rails 1 and 2. Totals: **rail 1: 4, rail 2: 7, rail 3: 6, rail 4: 6, rail 5: 3** (sum 26).

(b) First rail: $\lceil n / (2(d-1)) \rceil$. Last rail: $\lfloor (n + d - 2) / (2(d-1)) \rfloor$. Middle rails get two letters per full period.

(c) Rail 2 has the most letters (7); rails 3 and 4 have 6 each.

Solution RF4

(a) Undo in reverse order. Step 1: un-rail with depth 3 → HQHPBDWWDFNVDWGDZQ. Step 2: subtract 3 from every letter → **ENEMYATTACKSATDAWN**.

(b) For this pair, no: a shift acts letter by letter and a rail fence only moves letters, so they commute. (A substitution that depends on the position, such as Vigenère, would not commute with a transposition.)

(c) Caesar: confusion (weak). Rail fence: diffusion (positions). Together they form a very weak product cipher; DES iterates the idea sixteen times with strong S-boxes.

Solution RF5

(a) When d ≥ n the zig-zag never turns: the text is written straight down the rails, so **d ≥ 12** gives the plaintext. For d = 11 the zig-zag turns once and gives ABCDEFGHIJLK, which is different.

(b) Compute each depth: d=2: ACEGIKBDFHJL; d=3: AEIBDFHJLCGK; d=4: AGBFHLCEIKDJ; d=5: AIBHJCGKDFLE; d=6: AKBJLCIDHEGF; d=7: ABLCKDJEIFHG; d=8: ABCDLEKFJGIH; d=9: ABCDEFLGKHJI; d=10: ABCDEFGHLIKJ; d=11: ABCDEFGHIJLK; d=12: ABCDEFGHIJKL. Number of distinct strings: **11** (d = 12 equals the plaintext, which is also what d ≥ 12 gives).

(c) The key space grows only linearly with n (about n − 1 depths), so the rail fence has a tiny key space and can be brute-forced by hand.

9. One-Time Pad (Vernam)

Solution OT1

(a) $C_1 - C_2 \equiv P_1 - P_2 \pmod{26}$ because the key cancels: LQCAXAVWFFAJ.

(b) Key = $C_1 - P_1$ for known positions: positions 1–4: XQZJ; positions 10–12: WPD. Compare with the true key XQZJMBRKTWPD: matches. ✓

(c) Apply the recovered key letters to C_2: positions 1–4: **HOLD**, positions 10–12: **ION**. Pattern "HOLD…ION" suggests **HOLDPOSITION** (full key then confirms it).

(d) Never reuse an OTP key. Two ciphertexts leak $P_1 - P_2$, which crib-dragging can unwind; the cipher's perfect secrecy relies absolutely on single use.

Solution OT2

(a) The digit is at byte position 5 (zero-based 4). '1' = 0x31, '9' = 0x39. Required XOR delta = 0x31 ⊕ 0x39 = **0x08**.

(b) Flip byte 5 of C: 39 ⊕ 08 = 31. New ciphertext: **6A8427B131E452**.

(c) P' = C' ⊕ K = (C ⊕ Δ) ⊕ K = P ⊕ Δ. Byte 5: 0x31 ⊕ 0x08 = 0x39 = '9'. Decrypted: **PAY 900**.

(d) The OTP gives perfect secrecy (confidentiality) but **no integrity**: an attacker who knows the plaintext structure can modify it predictably without knowing the key. A MAC is needed on top.

Solution OT3

(a) Uniform key: P(C = A) = Σ P(m)·(1/3) = 1/3. P(M = A, C = A) = P(A)·P(K=0) = 0.5/3. Posterior = (0.5/3)/(1/3) = **0.5 = prior**. Perfect secrecy holds.

(b) C = 0 (A) arises from (M=A, K=0): 0.5·0.6 = 0.30; (M=B, K=2): 0.3·0.2 = 0.06; (M=C, K=1): 0.2·0.2 = 0.04. Total = 0.40. Posteriors: P(A|A) = 0.30/0.40 = **0.75**, P(B|A) = 0.06/0.40 = **0.15**, P(C|A) = 0.04/0.40 = **0.10**.

(c) The posterior differs from the prior (0.5, 0.3, 0.2), so the ciphertext leaks information: seeing A raises the belief that the message is A from 50% to 75%. A **non-uniform key destroys perfect secrecy**. The key must be uniformly random.

Solution OT4

(a) 2×10^6 / 8 = 250,000 bytes/s × 5,400 s = **1,350,000,000 bytes ≈ 1.35 GB**.

(b) 128 GB / 1.35 GB ≈ **94 complete calls**. Then the pad is exhausted.

(c) The key must be generated, distributed securely (another channel as secure as the data) and never reused; the storage and logistics cost equals the data volume. That is the key-distribution problem.

Solution OT5

(a) Key = C − P letter by letter: E−A = E, Y−R = H, S−E = O, X−N = K, G−A = G. **Key = EHOKG**. Check: ARENA + that key = EYSXG.

(b) For every 5-letter string there is exactly one key that maps the ciphertext to it, and every key is equally likely, so the ciphertext carries no information about which plaintext was sent (perfect secrecy). An interrogator cannot prove which plaintext is real.

(c) In the Caesar case (EVIRE) the brute force gave two meaningful words only because the key space is just 26. For the OTP, **every** plaintext of that length is possible and every one of them is equally plausible.

10. DES, 2DES and 3DES

Solution DE1

(a) $E(R_1)$ = **011101 011110 101001 010100 001100 001010 101000 001001**.

(b) $E(R_1) \oplus K_2$ = **000011 000100 010010 001101 111010 110110 001111 101100**.

(c) S-box lookups (outer bits = row, middle four = column): S1(000011) row 1, col 1 = 15, S2(000100) row 0, col 2 = 8, S3(010010) row 0, col 9 = 13, S4(001101) row 1, col 6 = 0, S5(111010) row 2, col 13 = 3, S6(110110) row 2, col 11 = 10, S7(001111) row 1, col 7 = 10, S8(101100) row 2, col 6 = 14. Output: **1111 1000 1101 0000 0011 1010 1010 1110**.

(d) P-box: f = **0011 1100 1010 1011 1000 0111 1010 0011**. $R_2 = L_1 \oplus f$ = **1100 1100 0000 0001 0111 0111 0000 1001**. $L_2 = R_1$ = **1110 1111 0100 1010 0110 0101 0100 0100**.

Solution DE2

(a) The schedule 1, 1, 2, 2, ...: round 1 shifts by 1, round 2 by 1. After round 2 the halves have been shifted by a total of 2.

(b) $C_2 = C_0$ rotated left by 2: **1100001100110010101010111111**. D_2 = **0101010110011001111000111101**.

(c) Concatenate C_2D_2 (56 bits) and pick the 48 bits listed in PC-2: K_2 = **011110 011010 111011 011001 110110 111100 100111 100101**.

(d) PC-2 drops positions 9, 18, 22, 25, 35, 38, 43, 54 of the 56-bit register C_iD_i (the numbers missing from the PC-2 table). Because the register is rotated between rounds, a different key bit sits in each dropped position every round, so **no key bit is unused overall**: each of the 56 key bits appears in about 14 of the 16 subkeys.

Solution DE3

(a) S1(101011): row 3, col 5 → 9 = 1001, S2(000111): row 1, col 3 → 7 = 0111, S3(010010): row 0, col 9 → 13 = 1101, S4(111100): row 2, col 14 → 8 = 1000, S5(001001): row 1, col 4 → 4 = 0100, S6(110110): row 2, col 11 → 10 = 1010, S7(100101): row 3, col 2 → 13 = 1101, S8(011010): row 0, col 13 → 0 = 0000.

(b) 32 bits: **1001 0111 1101 1000 0100 1010 1101 0000**.

(c) P-box: **0101 0110 1011 0101 0100 0001 0101 0011**.

(d) Each S-box has 4 rows, 16 columns and 4-bit entries (0–15); 8 S-boxes in total take 48 bits in and give 32 bits out.

Solution DE4

(a) Round 1: f(0110, 1100) = (0100) ⊕ (1100) = 1000. R_1 = 1011 ⊕ 1000 = 0011, L_1 = 0110. Round 2: f(0011, 0101) = 0111. $R_2 = L_1 \oplus f = 0001$, L_2 = 0011. After swap: **ciphertext = 0001 0011**.

(b) Feed the ciphertext (L'_0, R'_0) = (0001, 0011) into the same network with the subkeys reversed. Round 1 (K_2): L'_1 = 0011, R'_1 = 0001 ⊕ f(0011, 0101) = 0001 ⊕ 0111 = 0110. Round 2 (K_1): L'_2 = 0110, R'_2 = 0011 ⊕ f(0110, 1100) = 0011 ⊕ 1000 = 1011. Swap the halves: **$(1011, 0110) = (L_0, R_0) = (1011, 0110)$**, the original block. ✓

(c) For a key such as K = 1111, f(0000, K) = (0000 AND 1111) ⊕ rotl(0000) = 0000 and f(1111, K) = (1111 AND 1111) ⊕ rotl(1111) = 1111 ⊕ 1111 = 0000: two different inputs give the same output, so f is many-to-one and cannot be inverted. Decryption never inverts f: it recomputes f(R, K) from the half it already has and XORs it again, because (X ⊕ f) ⊕ f = X. That is why DES S-boxes (6 bits in, 4 bits out) may be non-invertible.

Solution DE5

(a) **Key schedule:** PC-1, the shifts and PC-2 only select and move bits, so the subkeys of ~K are the complements of the subkeys of K. **Round function:** the expansion only copies bits, so E(~R) = ~E(R). Then ~E(R) ⊕ $\sim K_i$ = E(R) ⊕ K_i (two complements cancel in an XOR), so the 48 bits entering the S-boxes are **identical**, and f(~R, $\sim K_i$) = f(R, K_i). **Induction:** if (L, R) become (~L, ~R), the new right half is ~L ⊕ f(R, K) = ~(L ⊕ f(R, K)), so both halves stay complemented in every round. IP and FP are permutations and commute with complementation. Hence the output is ~C.

(b) ECCBA8866443200E is the bitwise complement of 133457799BBCDFF1 (flip every hex digit: 1→E, 3→C, 4→B, 5→A, 7→8, 9→6, B→4, C→3, D→2, F→0), and FEDCBA9876543210 is the complement of 0123456789ABCDEF. By the property the ciphertext is the complement of 85E813540F0AB405: **7A17ECABF0F54BFA**. (Checked with a DES implementation.)

(c) With a chosen-plaintext attack the attacker obtains E_K(P) and E_K(~P) for the unknown key. Testing a candidate key K′ on P against the first ciphertext also tests ~K′ against the complement of the second ciphertext. Only half of the keys (2^{55}) need to be tried, so the effective brute-force work drops from 2^{56} to 2^{55}.

Solution DE6

(a) 8 × 8 = **64** key pairs.

(b) Forward table, k_1 → $E_k_1(7)$: 0→5, 1→6, 2→7, 3→8, 4→9, 5→10, 6→11, 7→12. Backward table, k_2 → $D_k_2(0)$ (solve $3x + k_2 \equiv 0$, $3^{-1} = 11 \bmod 16$): 0→0, 1→5, 2→10, 3→15, 4→4, 5→9, 6→14, 7→3.

(c) Matching values: (k_1 = 0, k_2 = 1) at 5, (k_1 = 4, k_2 = 5) at 9, (k_1 = 5, k_2 = 2) at 10.

(d) Test the candidates on the second pair (2→3): (0,1): $E_1(E_0(2)) = 3$, (4,5): $E_5(E_4(2)) = 3$, (5,2): $E_2(E_5(2)) = 3$. All 3 candidates survive, so **the key pair is not unique**. Reason: $E_k_2(E_k_1(x)) = 3(3x + k_1) + k_2 = 9x + (3k_1 + k_2) \bmod 16$, and every surviving pair gives $3k_1 + k_2 \equiv 1 \pmod{16}$, i.e. the same composite function. Composition of affine maps is affine (a group), so the two keys collapse into one effective key, the same lesson as double affine and double shift. A real attack on a non-group cipher such as DES needs only a few extra pairs to remove false positives.

(e) Brute force: 64 double encryptions. MITM: 8 + 8 = 16 single operations plus a few verifications. For DES this is $2 \cdot 2^{56}$ operations instead of 2^{112}, the reason 2DES is only about 57-bit strong.

Solution DE7

(a) Average search = $2^{55} \approx 3.6\times10^{16}$ keys → $3.6\times10^{16} / 10^{9} = 3.6\times10^{7}$ s ≈ **1.1 years**. (Full space 2^{56}: 2.3 years.)

(b) Operations ≈ $2^{57} = 1.44\times10^{17}$ → 1.44×10^{8} s ≈ **4.6 years**. Memory = 2^{56} entries × 8 bytes = 2^{59} = **576,460,752,303,423,488 bytes ≈ 576 petabytes** (512 PiB). The memory requirement, not the time, is the practical barrier.

(c) 3DES with three independent keys: MITM on the middle needs about $2^{112} = 5.2\times10^{33}$ operations → 5.2×10^{24} s ≈ **1.6×10^{17} years**.

(d) In 2DES the composite key is 112 bits but MITM cuts it to 2^{57}. In two-key 3DES (K_1, K_2, K_1) there are three encryptions: the best known generic MITM-style attacks still require far more work or memory than 2DES (NIST: about 80-bit security), so it is stronger, but it is deprecated because of its 64-bit block (Sweet32) and low throughput.

Solution DE8

(a) M = 00000001 00100011 01000101 01100111 10001001 10101011 11001101 11101111. IP output = 11001100 00000000 11001100 11111111 11110000 10101010 11110000 10101010. **L_0 = CC00CCFF, R_0 = F0AAF0AA**.

(b) IP table first entry is 58 and last entry is 7: output bit 1 = input bit 58; output bit 64 = input bit 7.

(c) FP(IP(M)) = **0123456789ABCDEF** = M. ✓ (FP = IP^{-1}.)

(d) Both are fixed, public, key-independent permutations. An attacker simply applies the inverse. They were included for hardware loading convenience; they add no security.